

Software design Group Project

Prepared by

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1- Use Case Description

<i>Use Case1: Monitoring the Green Zone System</i>	
Description	This use case involves Managing and viewing the green zone system
Actors	Green Zone Operators
Related Use case(include)	
<i>Use Case1.1: View Vehicle Fines</i>	
Description	This use case allows the green zone operator to view fines issued for unauthorized green zone access
Actors	Green Zone Operators
<i>Use Case1.2: View Monitoring Points</i>	
Description	This use case allows operators to view information about the monitoring points within the green zone.
Actors	Green Zone Operators

Use Case2: Access Green Zone Permits and Accounts

Description	This use case involves issuing a green zone permit to vehicle owners who apply for access to the green zone
Actors	Green Zone Operators, Owner of Vehicle, Green Zone Resident

Related Use case(include)

Use Case2.1: View Permits Information

Description	This use case allows users to view information about their green zone permits.
Actors	Green Zone Operators, Owner of Vehicle, Green Zone Resident

Use Case2.2: View Account Information

Description	This use case allows users and the green zone operators to view the account information.
Actors	Green Zone Operators, Owner of Vehicle, Green Zone Resident

Use Case2.3: View Zone Trips Information

Description	This use case allows users and the green zone operators to view information about the trips within the green zone, including entry and exit times
Actors	Green Zone Operators, Owner of Vehicle, Green Zone Resident

Use Case3: Apply for Green Zone Resident Permit

Description	This use case allows green zone resident to apply for resident permits, enabling unrestricted access to the green zone.
Actors	Green Zone Operators, Green Zone Resident

Use Case4: Issue Green Zone Permit

description	This use case involves issuing permits to vehicle owners, allowing access to the green zone with appropriate billing.
Actors	Green Zone Operators, Owner of Vehicle

Use Case5: Detect Vehicles on Entry

description	This use case detects the presence of a vehicle entering the green zone.
Actors	Entry Vehicle Location Sensor, Vehicle Owner, Green Zone Resident

Use Case6: Scan Green Zone Permit on Entry

description	This use case scans the barcode permit of a vehicle entering the green zone
Actors	Entry Remote Laser Scanner, Vehicle Owner, Green Zone Resident

Use Case7: Enter Green Zone

description	This use case logs the entry of a vehicle into the green zone and updates the trip record in the system.
Actors	Owner of Vehicle, Green Zone Resident

Use Case8: Detect Unauthorized Vehicles

description	This use case captures and processes unauthorized vehicles entering the green zone without a valid permit Automatically
Actors	Vehicle Camera, External Vehicle License Database
Related Use case(include)	

Use Case8.2: Issue Fine

description	This use case issues a fine to the owner of the unauthorized vehicle entering the green zone.
Actors	Vehicle Camera, External Vehicle License Database

Use Case9: Detect Vehicles on Exit

description	This use case detects the presence of a vehicle exiting the green zone.
Actors	Exit Vehicle Location Sensor, Vehicle Owner, Green Zone Resident

Use Case10: Scan Green Zone Permit on Exit

description	This use case scans the barcode permit of a vehicle exiting the green zone.
Actors	Exit Remote Laser Scanner, Vehicle Owner, Green Zone Resident

Use Case11: Exit the Green Zone

Description	This use case logs the exit of a vehicle from the green zone and updates the trip record in the system.
Actors	Owner of Vehicle, Green Zone Resident

Use Case12: Invoice Account Deduct

Description	This use case deducts the appropriate fee from the vehicle owner's prepaid account based on the duration and time of the trip automatically
Actors	Owner of Vehicle, Pre-Paid Green Permits Database
Related Use case(include)	

Use Case12.1: Maximum Charge

Description	Ensures that the amount deducted from the pre-paid account does not exceed the maximum charge for a single day automatically
Actors	Owner of Vehicle, Pre-Paid Green Permits Database
Related Use case(include)	

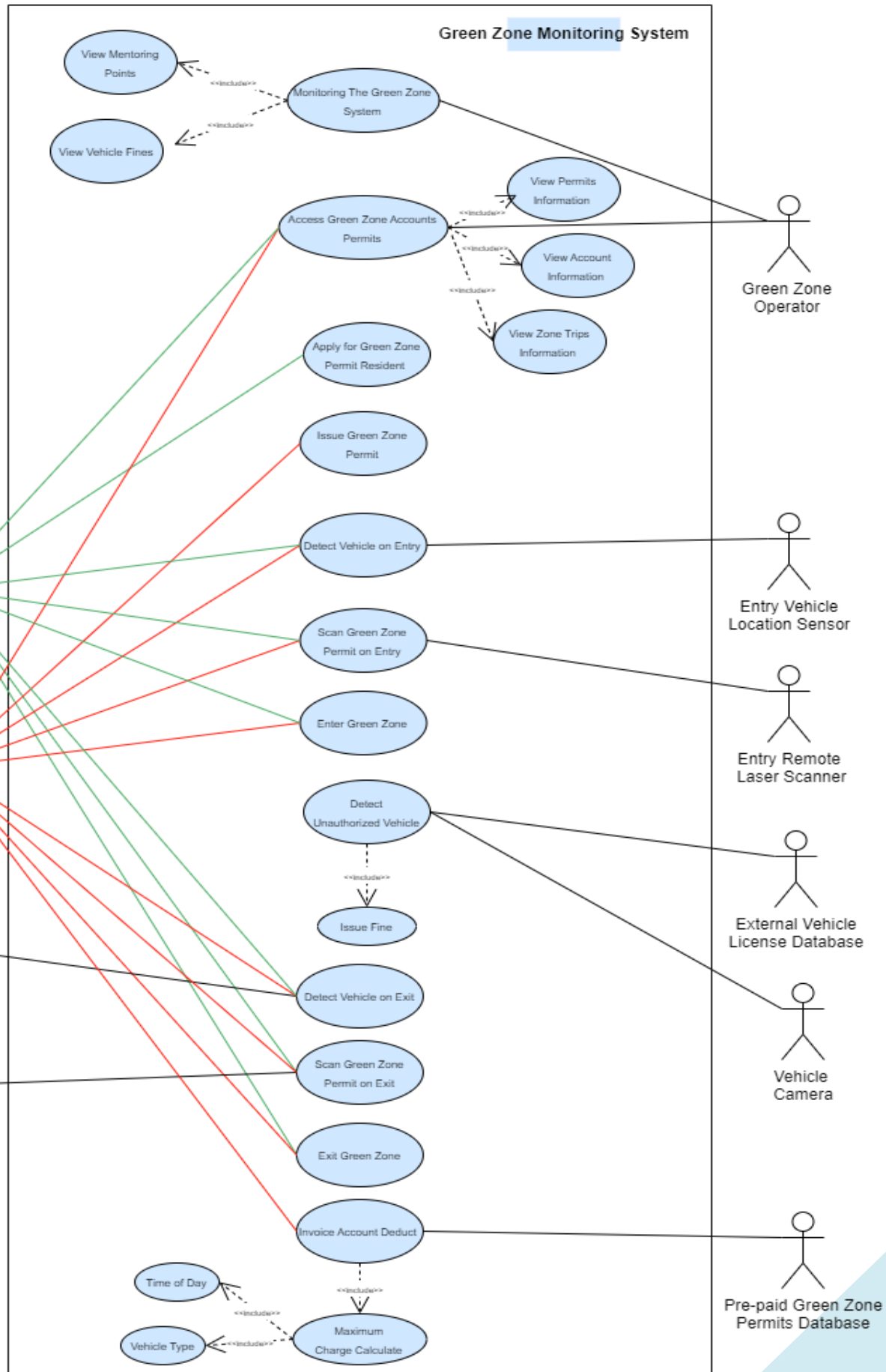
Use Case12.1.1.1: Time of The Day

Description	The system applies different rates based on the time of day when the vehicle is in the green zone
Actors	Owner of Vehicle, Pre-Paid Green Permits Database

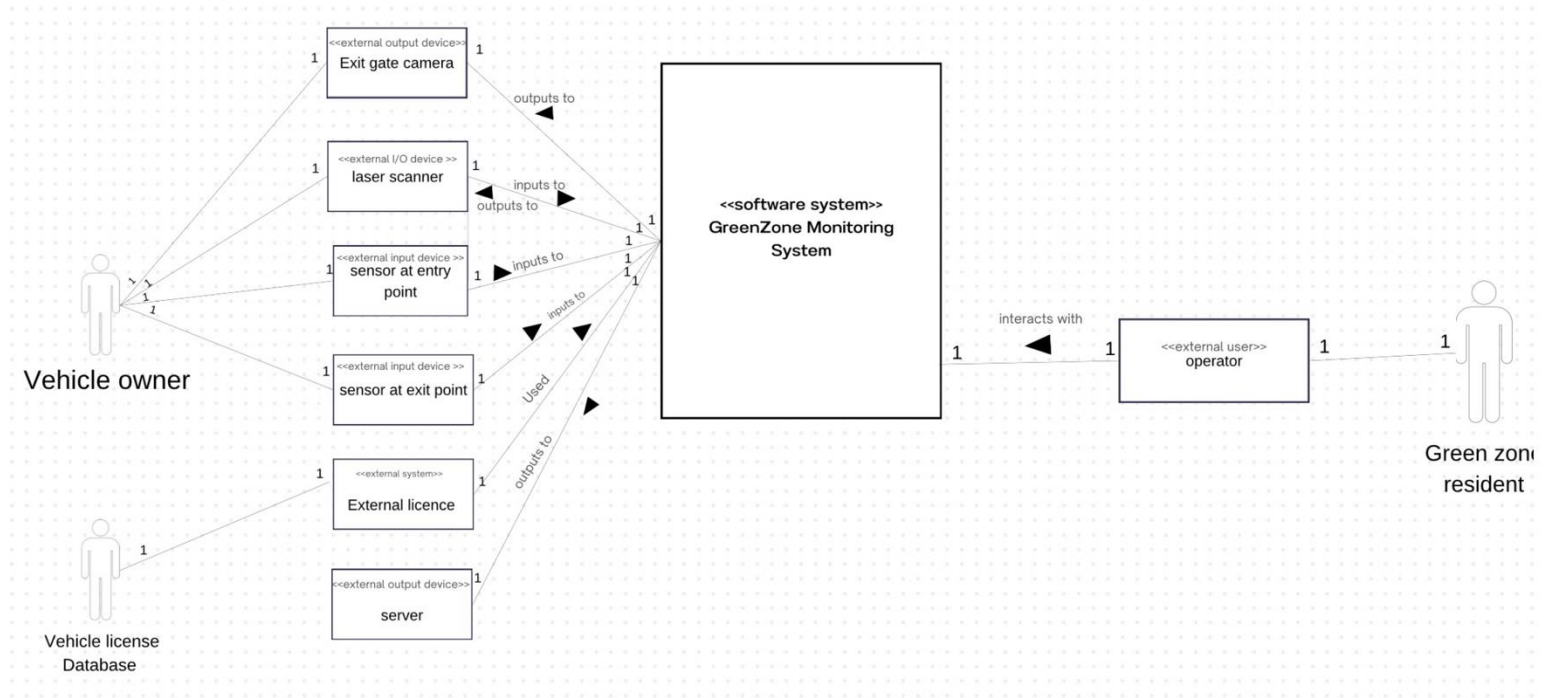
Use Case12.1.1.2: Vehicle Type

Description	The charge varies depending on the type of vehicle
Actors	Owner of Vehicle, Pre-Paid Green Permits Database

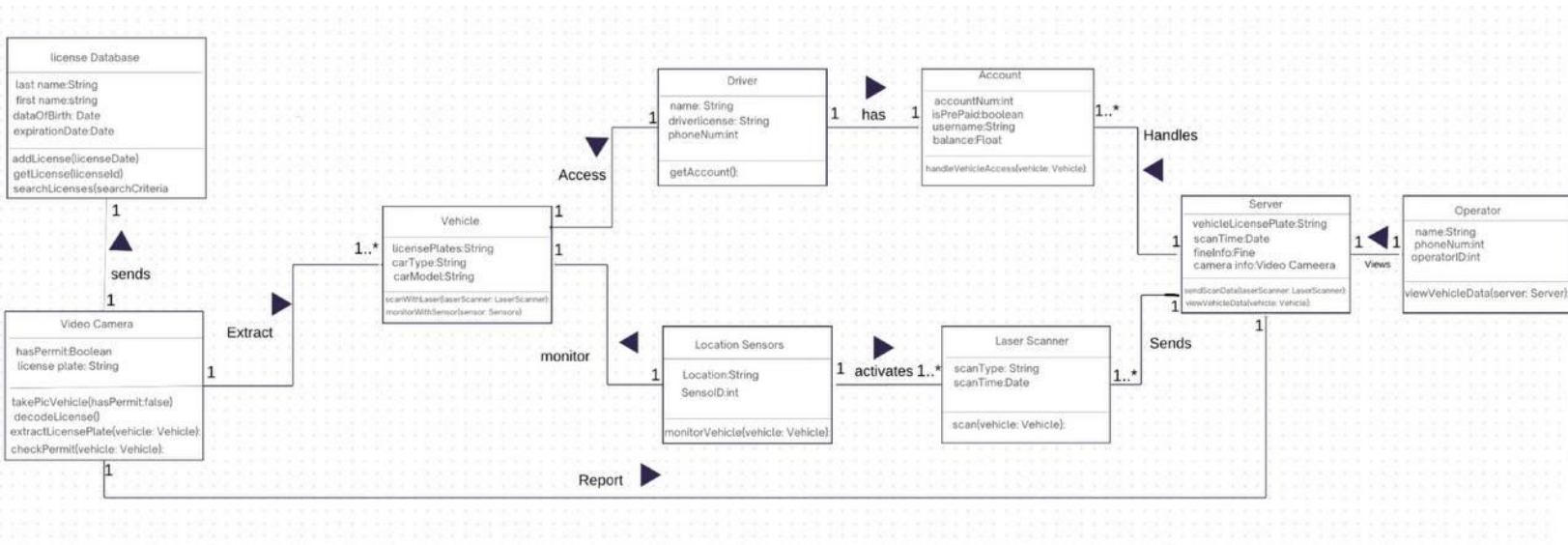
1- Use Case Diagram



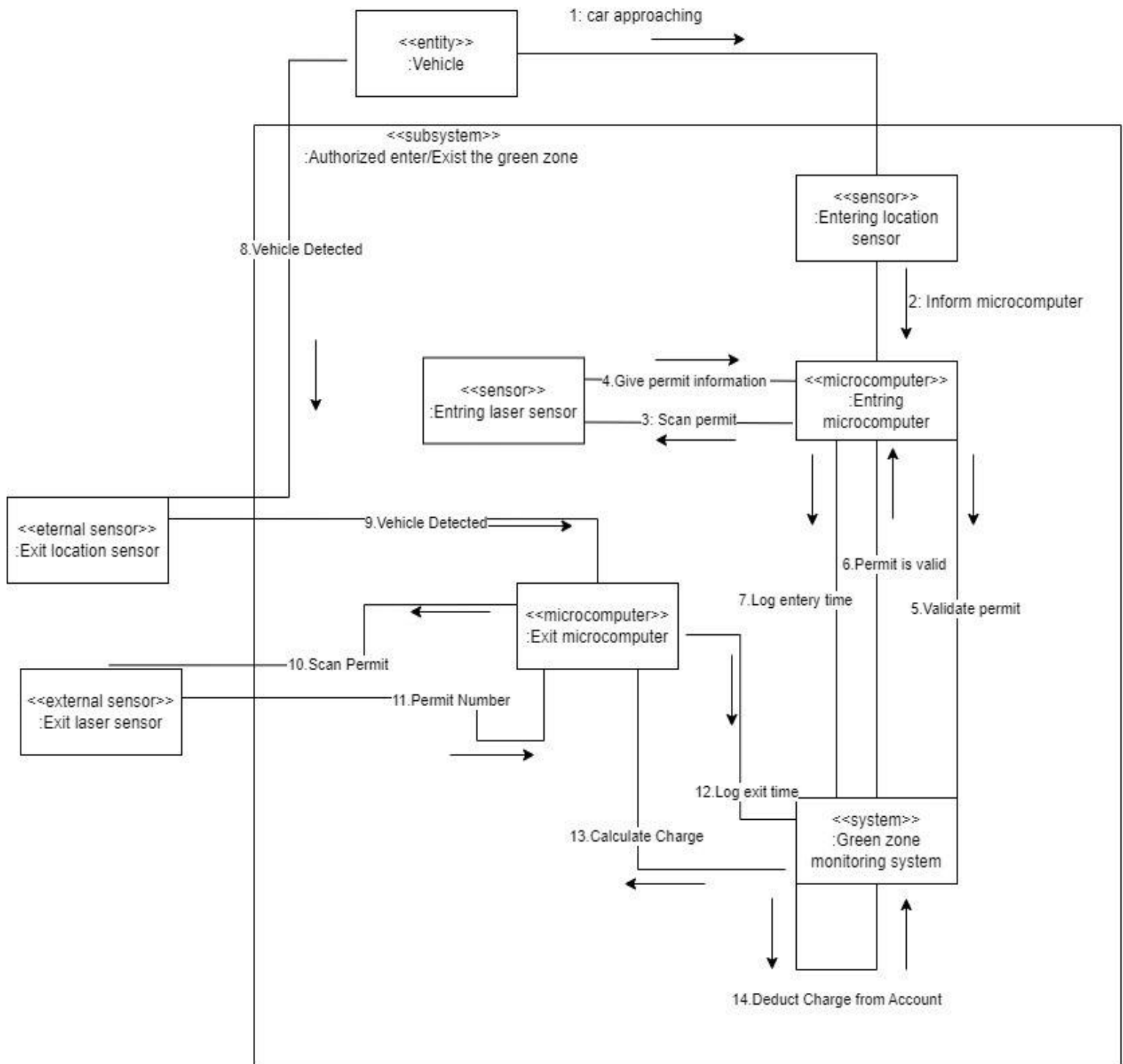
2- Context Class Diagram



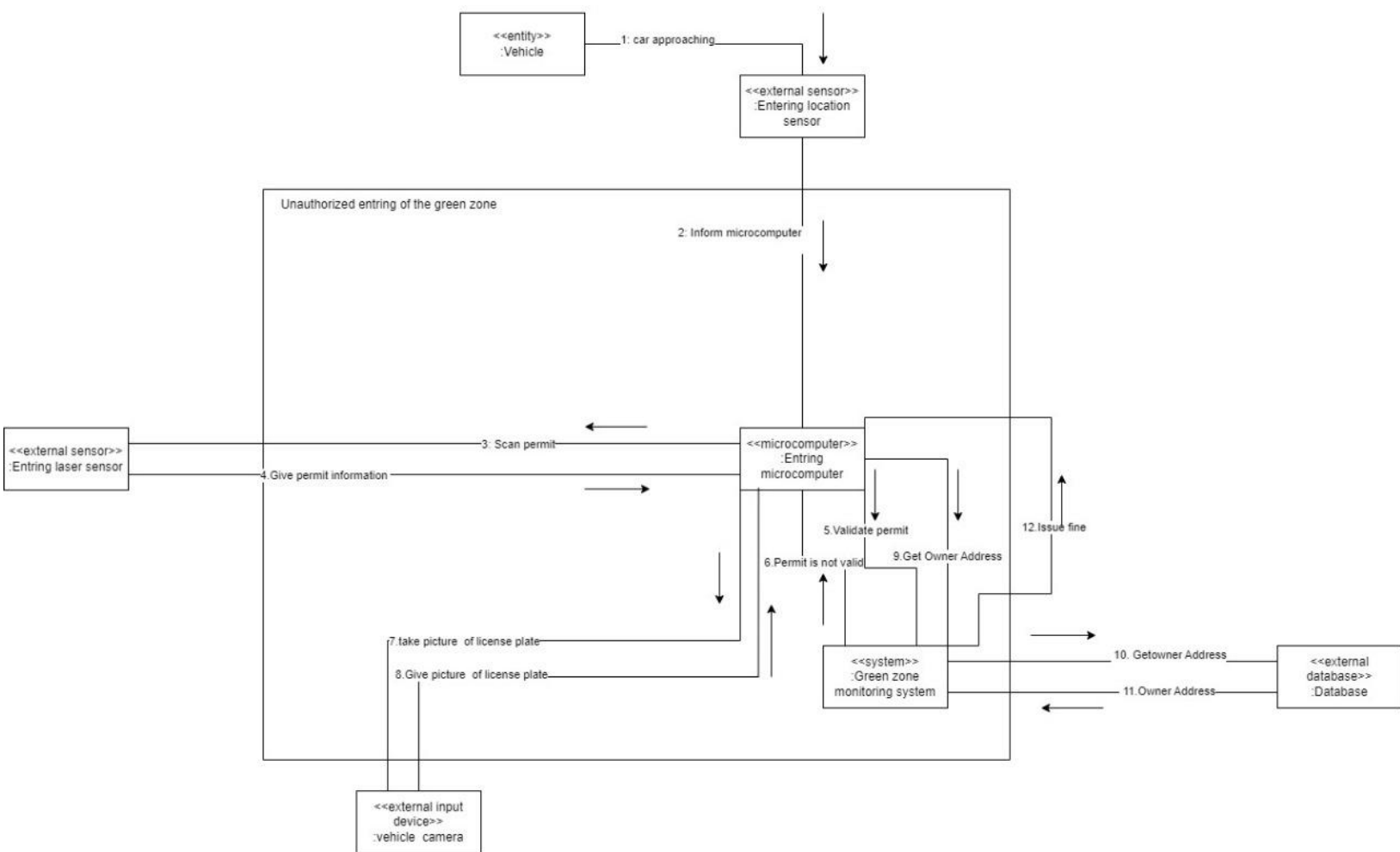
3- Static Diagram



4- Integrated Communication Diagram

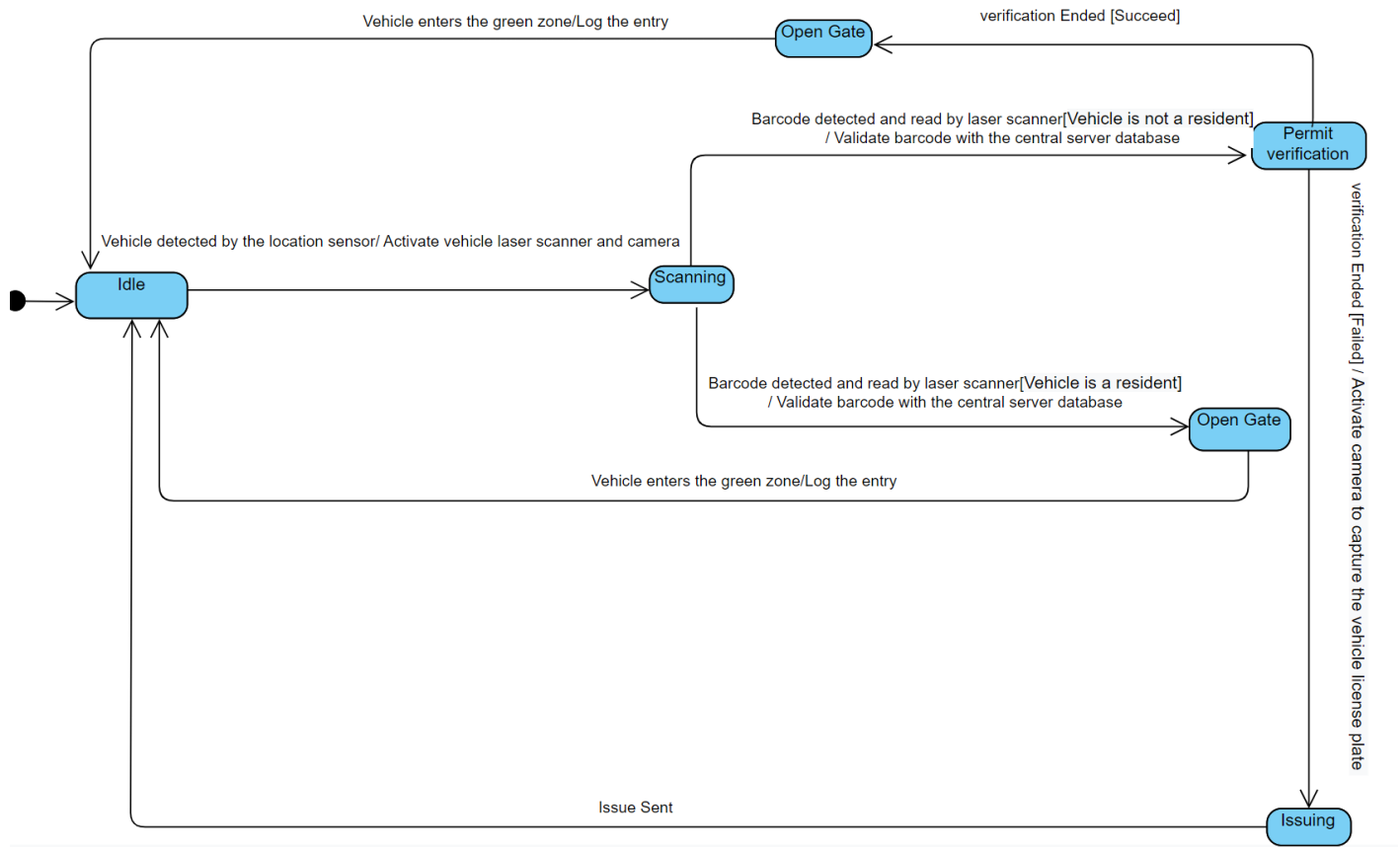


4.1. authorized Exit/Enter

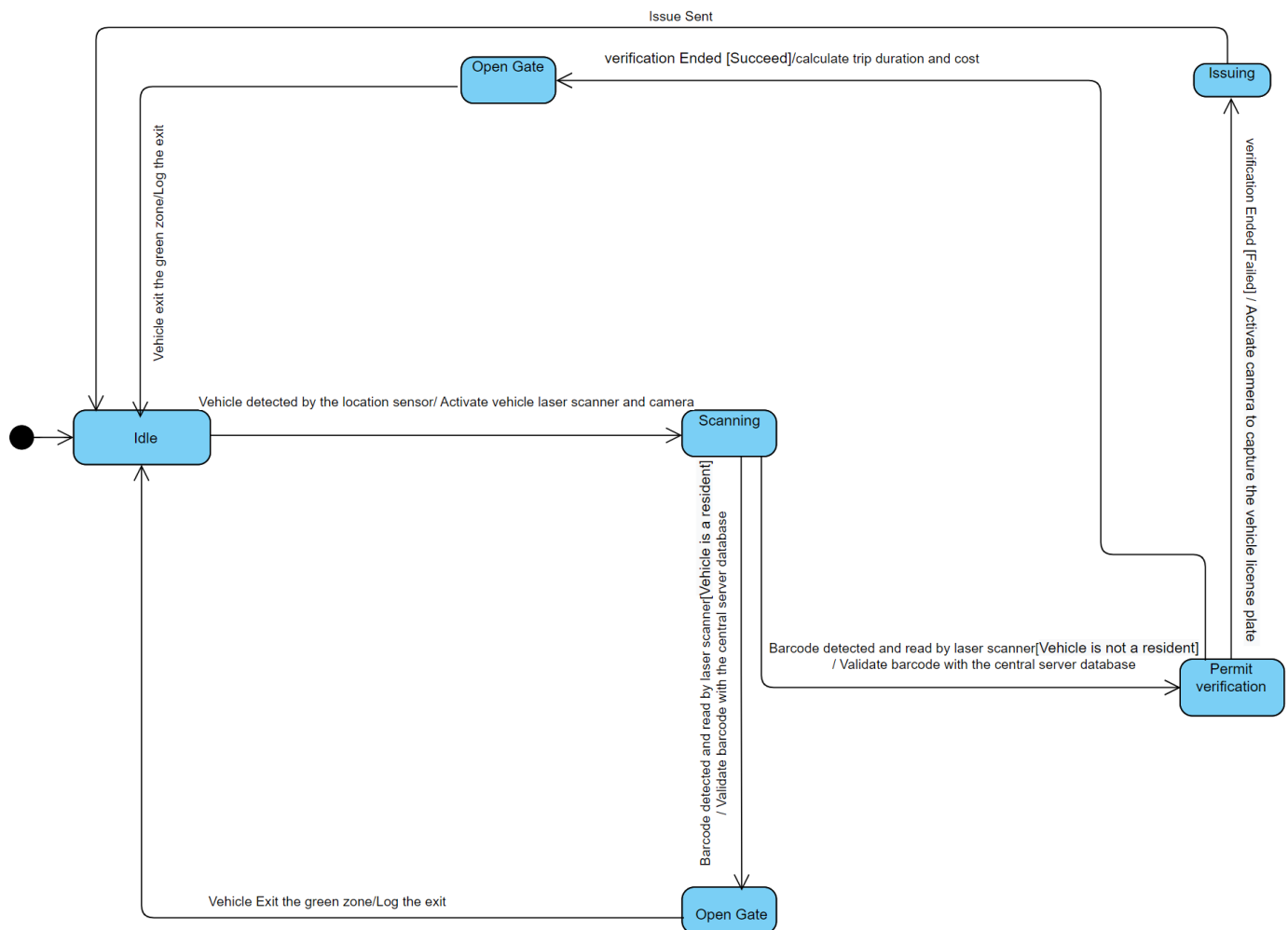


4.2. unauthorized Access

5- Statechart Diagram



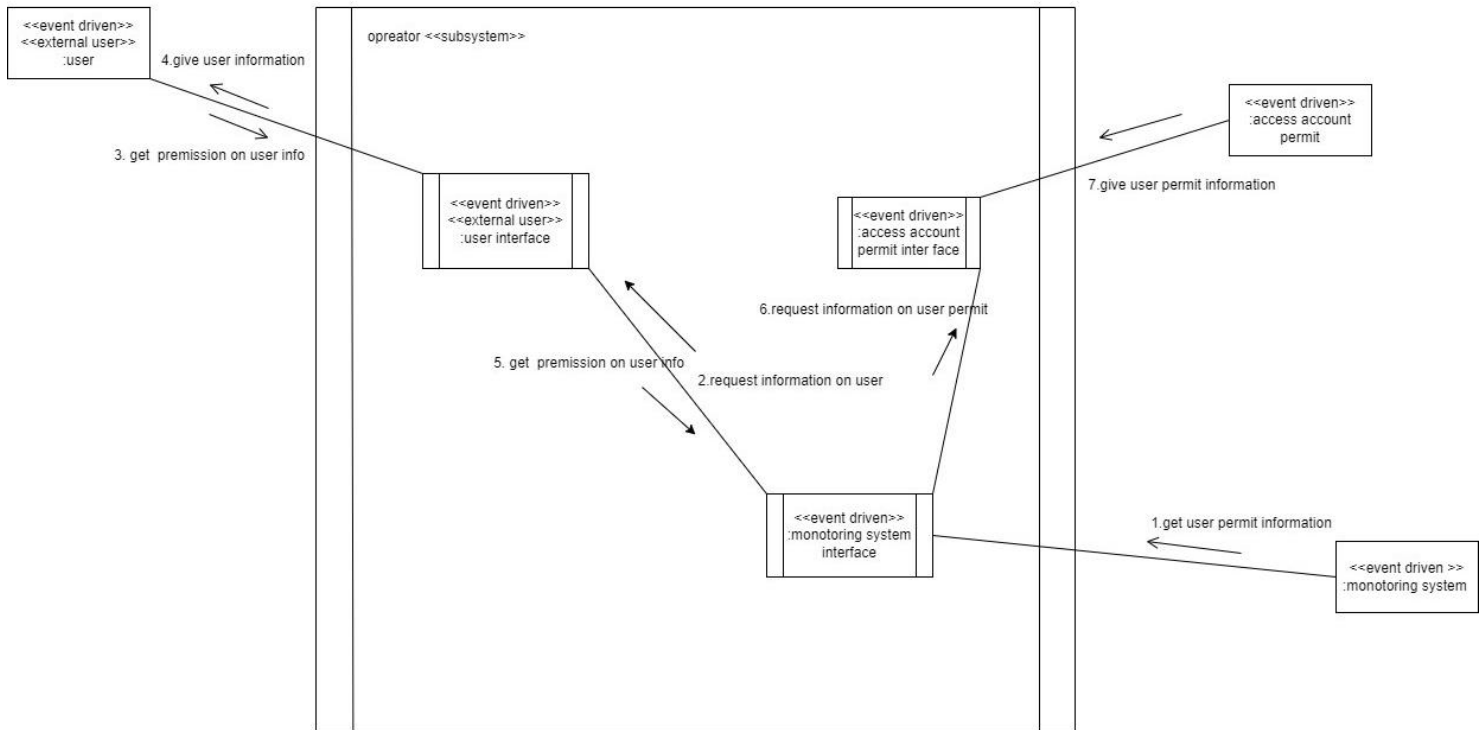
5.1. Entry Statechart



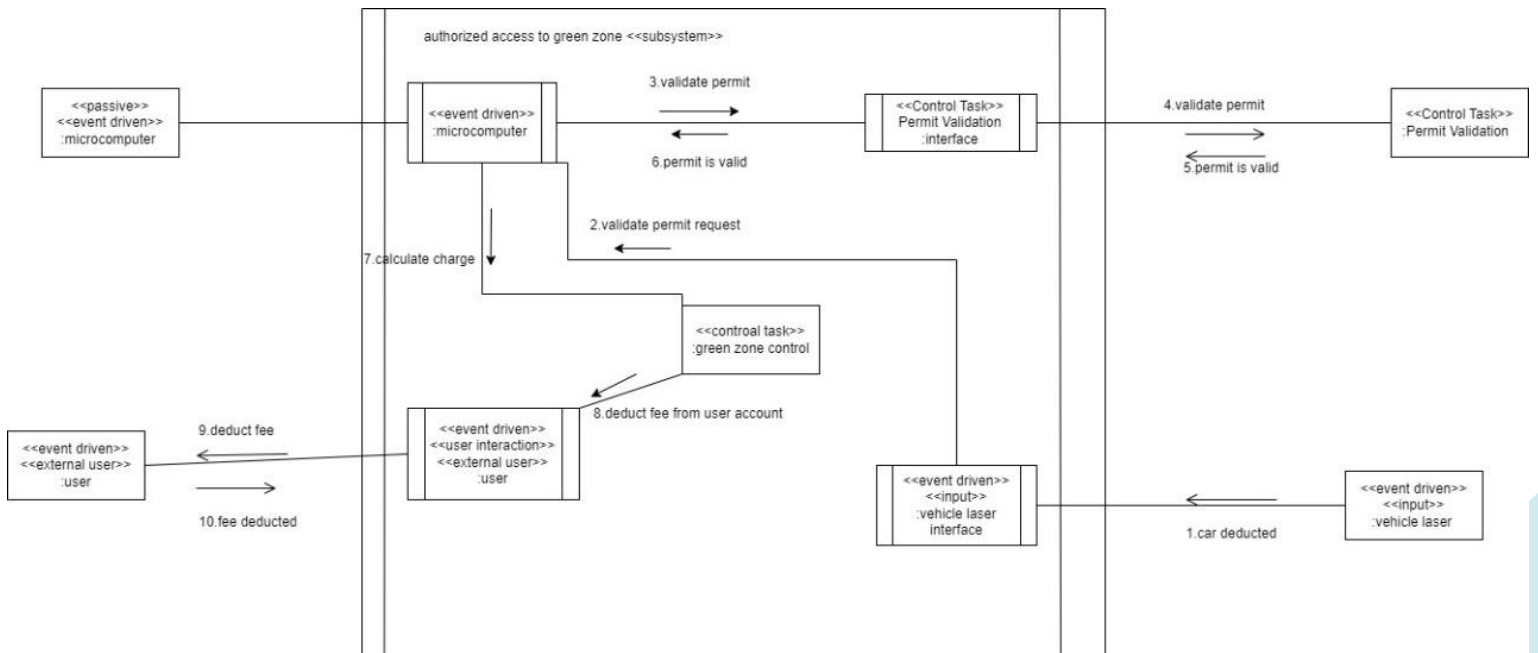
5.2. Exit Statechart

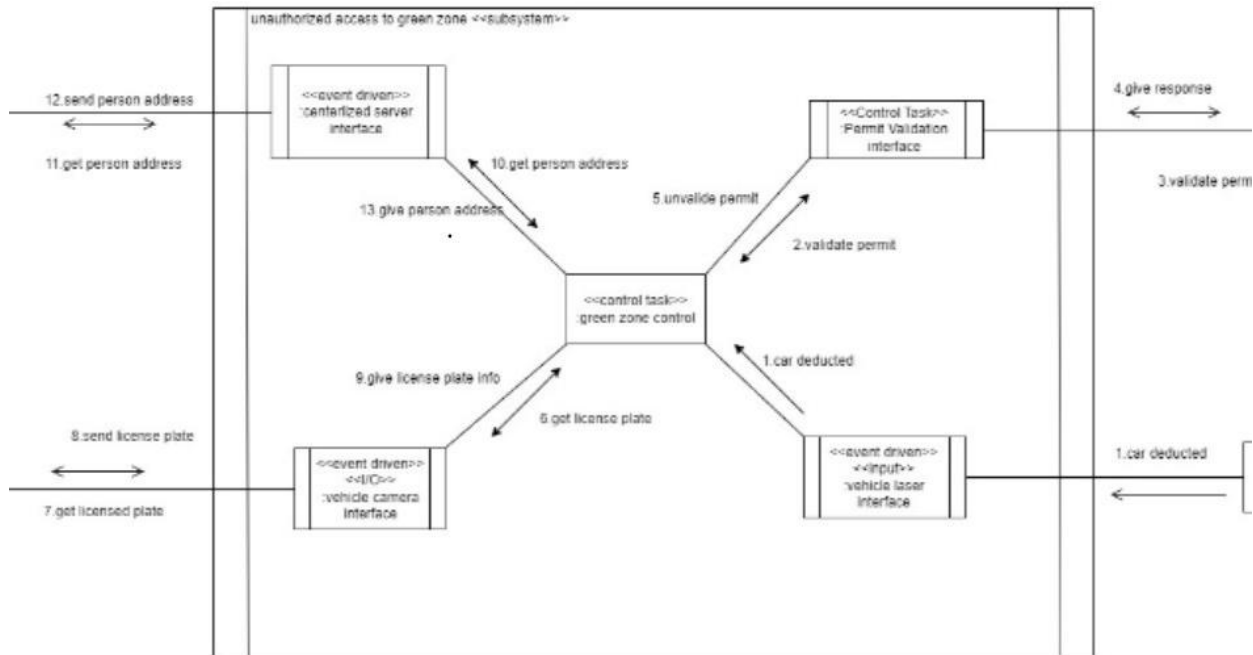
6- Task Architecture Diagram

6.1 operator management subsystem

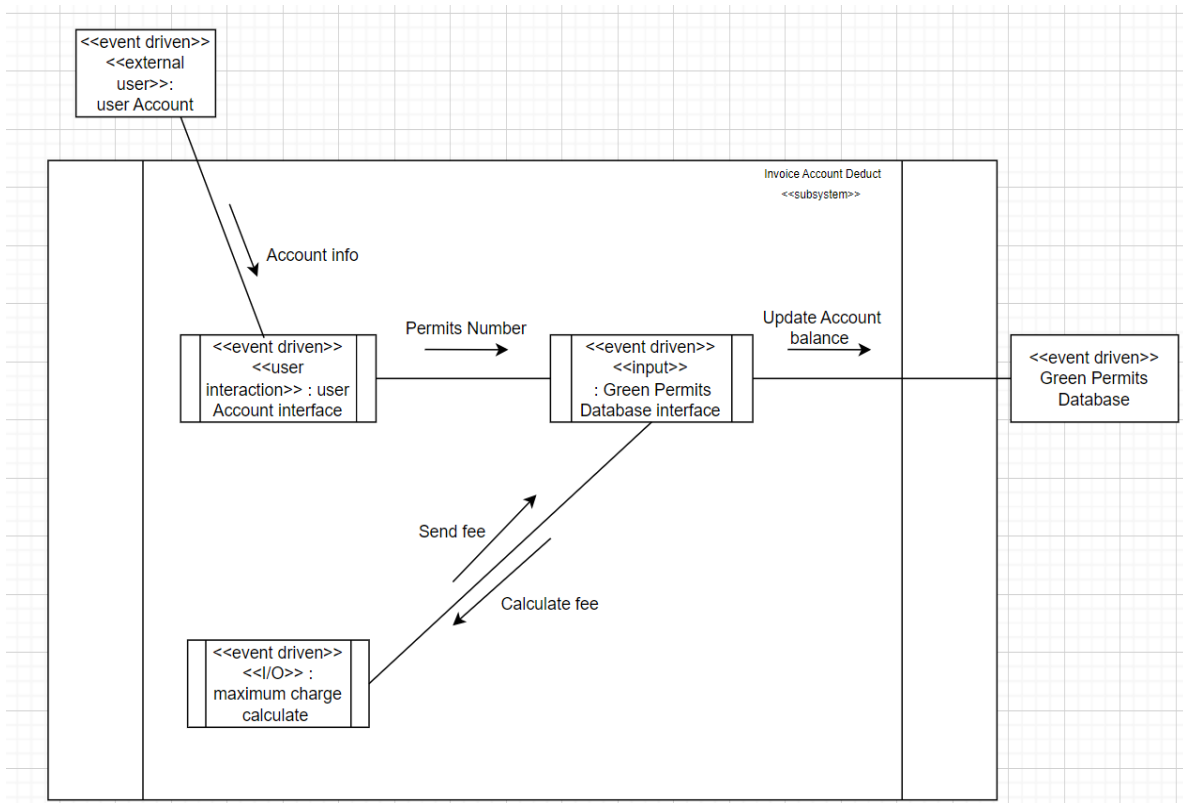


6.2 authorized Access subsystem





6.3 unauthorized Access subsystem



6.4 invoice account deduct subsystem